Md Erfanul Islam Bhuiyan

Backend & AI Engineer

eib.erfan@gmail.com | +49 175 9487977 | Berlin, Germany | erfanul.com | github.com/erfanul007
linkedin.com/in/erfanul007 | leetcode.com/u/erfanul007

Profile

Backend and AI-focused Software Engineer with **around 5 years** building enterprise SaaS platforms and event-driven distributed systems. I turn complex business requirements into reliable services in **Python and .NET**, build RAG and AI-agent workflows, and use AI across the software development lifecycle to speed up delivery.

Professional Experience

Senior Backend Developer Aug 2026 – Oct 2026
InfinitiBit *Dhaka, Bangladesh*

- Architected the **Python/FastAPI and PostgreSQL** backend of an AI VAT-validation product that checks invoices against German and EU tax law and cites the legal provision behind every verdict.
- Evolved the proof of concept into a multi-tenant SaaS MVP with SSO, role-based access, credit billing and audit logs, isolating each customer's data by PostgreSQL schema and database role on every query.
- Delivered tooling to create, monitor and sync each customer's tax-law knowledge graph, with versioning for easy rollback and human approval before any law change goes live.
- Defined a RabbitMQ event contract for the company's SaaS products and engineered Temporal workflows with a transactional outbox, sized to run **10,000** validations from one click and recover after restarts.
- Instrumented the API, workers and LLM calls end to end with OpenTelemetry and Grafana; set up Bifrost with DevOps as an AI gateway with per-app virtual keys to control token costs across LLM providers.
- Introduced a Claude Code delivery workflow; reviewed **35+** pull requests and mentored two junior developers.

Senior Software Engineer (Backend) Apr 2026 – Jul 2026
SDS Manager (Netpower) *Dhaka, Bangladesh*

- Created a .NET 10 document-intelligence service with smart chunking, sparse vectors and dense embeddings using Microsoft.Extensions.AI.
- Led development of a shared background-job system for quality-management microservices, with custom recurrence scheduling and OpenTelemetry tracing.
- Packaged RabbitMQ messaging into a reusable library for notification and email events between microservices, and set up CI/CD pipelines for multiple service repositories.

Senior Backend Developer Nov 2025 – Mar 2026
InfinitiBit *Dhaka, Bangladesh*

- Led backend development of a private GraphRAG chat system on a local GPU cluster, cutting response time from **10–12 minutes to under 2 minutes** with an asynchronous agent pipeline.
- Engineered prompts and context with cumulative history summarization for long conversations, and evaluated answer quality using LLM-as-a-judge.
- Created a document ingestion tool that converted **3 TB** of PDFs, Office files and images into text with metadata for the knowledge graph and model training.
- Developed a VAT audit agent with GraphBit in a 3-day hackathon that extracts rules from legal text and explains each finding.
- Architected the backend of a transfer-pricing tax platform, and delivered its single sign-on, role-based access, document storage and working-paper ingestion.
- Reviewed **45+** pull requests across the chat and tax platforms and planned work for two developers.

Backend Developer (.NET) Nov 2024 – Nov 2025
InfinitiBit *Dhaka, Bangladesh*

- Orchestrated AI analysis of bid documents (RFQ, technical, pricing) with background jobs, progress tracking and a circuit breaker.
- Built core modules of a .NET 8 bid-management platform for a leading German energy company as its largest contributor, with **150+** merged pull requests.
- Automated bid approval and e-signature with workflow rules, document version locking and approval reports; connected modules over RabbitMQ and integrated AWS and SharePoint.
- Designed the architecture and AWS cost model for a real-time power-utility data platform sized for about **7,500** sensor messages per second.

Software Engineer Jan 2022 – Jul 2024
Enosis Solutions *Dhaka, Bangladesh*

- Fixed high-priority production issues, including unauthorised data access, by building a central policy-based access control (PBAC) layer across APIs.
- Implemented Microsoft Entra SSO (OAuth2/OIDC) for a multi-tenant B2B SaaS platform with **100k+** users.
- Built a serverless, independently scaling email service that sends mail from clients' own accounts and templates.
- Migrated a hiring platform from Node.js to .NET 8 as a modular monolith, and a US dental practice system from VB.NET 3.5 to C# .NET 6.

Technical Skills

Languages — Python, C#, TypeScript, SQL, Go

AI & LLM — Agent Workflows, RAG, GraphRAG, Knowledge Graphs, LangGraph, MCP Servers & Tool Calling, Vector Search (Qdrant, pgvector), Local LLMs, CrewAI, LLM Gateways (LiteLLM, Bifrost)

Backend & Architecture — FastAPI, REST APIs, ASP.NET Core, Microservices, Event-Driven Systems, Domain-Driven Design, RabbitMQ, Temporal

Data & Cloud — PostgreSQL, SQL Server, Neo4j, Redis, AWS, Azure

DevOps & Observability — Docker, GitHub Actions, GitLab CI/CD, OpenTelemetry, Grafana (Prometheus, Tempo, Loki)

AI-Assisted Development — Claude Code, Agent Skills, AI-Driven SDLC

Personal Projects

Paceframe, *Goal and performance management platform* paceframe.erfanul.com

- Launched a web app for small organisations with goal tracking, continuous check-ins, weighted quarterly reviews and approval workflows.
- Designed the domain model and architecture; implemented the modular-monolith backend and React UI.

Technologies: C# .NET, PostgreSQL, React, TypeScript, Docker

CPorBit, *Personalised competitive programming mentor* github.com/smhimran/CPorBit

- Developed a Python/Django REST and PostgreSQL web app that analyses users' Codeforces submissions and suggests practice problems.
- Spearheaded system design and R&D, including the KNN-based problem-suggestion algorithm.

Technologies: Django REST, Python, PostgreSQL, React, TailwindCSS

Achievements

Employee of the Month, *InfinitiBit* Jul 2025
Awarded for contributions to backend and AI systems.

Competitive Programming

- Peak rank **Expert** on Codeforces (rating 1623): codeforces.com/profile/erfanul007.
- Solved **2500+** problems across online judges: linktr.ee/erfanul007.
- **Champion**, DIU Intra-University Programming Contest 2021.
- Competed in the ICPC Dhaka Regional 2019 and 2020 for Daffodil International University.

Education

BSc in Computer Science & Engineering Jan 2018 – Jan 2022
Daffodil International University *Dhaka, Bangladesh*
CGPA: **3.73/4.00**

Languages

English — C1 (IELTS Academic, Overall Band 7)

German — Basic (A1–A2), actively learning